

Vishva Arasan

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EDUCATION

University of California Santa Barbara

Bachelor of Science in Computer Science, Accelerated Track

Santa Barbara, CA

June 2027

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, HTML/CSS

Frameworks: Django, Django REST Framework, Flask, Node.js

Tools & Testing: KLEE, DeepSeek V3.2, Git, Postman, PostgreSQL, SQLite, Vim, VS Code, AWS Cloud Practitioner

Core Concepts: LLM Inference, Symbolic Execution, Fuzzing, Static Analysis, REST APIs, ORM

RESEARCH EXPERIENCE

Undergraduate Research Assistant

UCSB Verification Lab (Advisor: Prof. Bultan)

Sep 2025 – Present

Santa Barbara, CA

- DARPA HARDEN Challenge: Engineering STASE (Static Analysis Guided Symbolic Execution), a custom toolchain designed to detect Weird Machine (WMj) vulnerabilities within the Linux Kernel.
- Developing an automated pipeline using **DeepSeek V3.2** to generate C drivers for **KLEE**, automating **symbolic execution** for vulnerability detection.
- Implementing **guided reasoning agents** to detect error hotspots via **static analysis** and generate instrumented files with automated assert statements.
- Engineered a data abstraction layer converting C structures into **JSON representations** to facilitate **LLM reasoning** over program state.
- Evaluating detection **precision and recall** against the **Juliet test suite** to benchmark performance against the STASE tool.

PROJECTS

KOS Operating System Kernel | C, k-threads, Semaphores

Mar 2026

- Engineered an **operating system** from scratch in C implementing 15+ core system calls including process management (fork, execve, wait, exit) and file descriptor handling (dup, dup2, close).
- Architected a thread-safe, atomic inter-process communication (**IPC**) pipe system utilizing circular buffers and strict semaphore synchronization (**mutexes**) to guarantee data integrity across multiple processes.
- Managed process control blocks (**PCBS**) and file descriptor tables, utilizing strict reference counting for readers and writers to prevent memory leaks and ensure safe process blocking/unblocking.

Global Countries Management API | Python, Django, REST Framework

Dec 2025

- Developed a **RESTful API** implementing full **CRUD operations** and custom search logic via **Django QuerySets**.
- Standardized **JSON error handling** and data integrity through **ModelSerializers** and input validation logic.

Task Management API | Python, Flask, SQLAlchemy

Jan 2026

- Built a backend service using **Flask** and **SQLite**, utilizing **SQLAlchemy ORM** for normalized database schema management.
- Performed comprehensive endpoint testing via **Postman** to ensure compliance with **stateless REST** architectural standards.

PROFESSIONAL EXPERIENCE

Computer Science Intern

CUBUS Solutions Inc. (Merged with Tyfone)

April 2023 – July 2023

Livermore, CA

- Supported the development of digital banking services using **SQL databases** and **JavaScript APIs**.

High School Coding Tutor

Granada High School

Aug 2023 – June 2024

Livermore, CA

- Mentored students in **Python and Java**, debugging algorithmic issues and explaining AP Computer Science concepts.